

Isha Harish

New York City, NY | theishaharish@gmail.com | +1 (929) 377-9074 | linkedin.com/in/isha-harish | github.com/isha-harish

SUMMARY

Software and AI engineer with experience building backend services, LLM-based tools, research systems, and cloud applications using Python, C++, SQL, FastAPI, PyTorch, Docker, Kubernetes, and Google Cloud. Background includes agent systems, machine learning, computer vision, model evaluation, and production software development.

EDUCATION

- New York University**, M.S. in Computer Engineering Sept 2024 – May 2026
• **GPA:** 3.8/4.0 **Relevant Coursework:** Machine Learning, Deep Learning, Efficient AI, Distributed Systems, Computer Architecture, ML Cybersecurity, ML Finance
- Manipal Institute of Technology**, B.Tech in Mechatronics Engineering Oct 2020 – July 2024
• **GPA:** 8.06/10.0 **Minor:** Robotics and Automation

SKILLS

- **Languages:** Python, C++, SQL, Bash
- **Software Engineering:** Object-oriented design, data structures and algorithms, REST APIs, unit testing, test automation, debugging, Git, Linux
- **Machine Learning:** PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, fine-tuning, model evaluation, model optimization, data preprocessing, NumPy, Pandas
- **LLM and Agent Systems:** LLMs, RAG, embeddings, tool calling, structured outputs, MCP, LangChain, LangGraph, OpenAI-compatible APIs, agent evaluation
- **Backend and Cloud:** FastAPI, Flask, AsyncIO, WebSockets, Docker, Kubernetes, Google Cloud, Cloud Run, GitHub Actions, CI/CD
- **Data and Applied Research:** PostgreSQL, MongoDB, Cosmos DB, SQL Server, ChromaDB, vector search, computer vision, speech recognition, agent-based modeling, simulation, QGIS, GDAL, Shapely

EXPERIENCE

- AI Engineer Intern**, Check Point Software Technologies June 2025 – Dec 2025
• Built 3+ Python and FastAPI services for internal AI applications, including a web research agent, model-routing service, retrieval workflows, and tool-calling endpoints.
• Developed retrieval-based answer generation with source citations, structured JSON outputs, model routing, and LangChain/LangGraph orchestration across multiple LLM providers.
• Implemented an OpenAI-compatible /v1/chat/completions proxy and MCP integrations supporting HTTP streaming, SSE, and stdio tools; containerized and deployed services to Google Cloud Run using Docker and GitHub Actions.
- Data Science Research Intern**, Indian Institute of Science, CDS Jan 2024 – June 2024
• Developed a Python and Mesa agent-based simulation to study human-crocodile conflict, modeling crocodile movement, human activity, and intervention strategies across 500+ scenario runs.
• Built geospatial data pipelines using QGIS, GDAL/rasterio, Shapely, OSMnx, and KDE to generate trajectories, risk maps, and conflict hotspot analyses.
• Evaluated mitigation strategies through simulation experiments and developed a decision-support system that identified interventions associated with an approximately 20
- Machine Learning Research Intern**, University of Washington Aug 2023 – Oct 2023
• Developed NLP pipelines for misinformation detection, comparing baseline machine learning models and engineered linguistic, lexical, and text-based features through cross-validation, error analysis, and performance evaluation.
- Computer Vision Engineer Member**, NYU RoboMaster Feb 2026 – May 2026
• Developed and deployed a real-time YOLOv8 perception pipeline on NVIDIA Jetson for target detection, tracking, and angle estimation; profiled inference latency and debugged CPU-GPU data transfers to improve runtime performance.

RESEARCH & TECHNICAL PROJECTS

- **Smart Receipt Processing System | Qualcomm Hackathon** – Developed a FastAPI document-processing pipeline using Donut and LLaMA to extract merchant, date, line-item, tax, and total fields from noisy receipt images into validated JSON outputs.
- **AG News Classification with LoRA** – Fine-tuned RoBERTa-base using Hugging Face Transformers and LoRA adapters on query and value projection layers, training 741K parameters and achieving 88.1
- **NYC Lens | Google Developer Hackathon, Top 10** – Developed a React and FastAPI application for landmark identification and question answering using image and text inputs, retrieval, and tool-using LLM workflows.
- **BetweenLines | Speech and NLP Assistant** – Built a speech-processing pipeline with Whisper transcription and a task-specific NLP model to classify sarcasm, passive aggression, tone, and conversational intent.